

Ultra-thin film turns smartwatch into health monitoring system

Researchers have developed a thin adhesive film that could upgrade a consumer smartwatch into a powerful health-monitoring system. The system looks for chemical indicators found in sweat to give a real-time snapshot of what's happening inside the body. According to the study, published in the journal of Science Advances, the researchers have engineered a disposable, double-sided film that attaches to the underside of a smartwatch.

The film can detect molecules such as metabolites and certain nutrients that are present in body sweat in very tiny

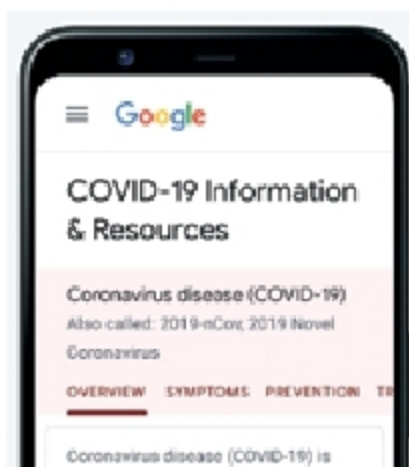
amounts. They also built a custom smartwatch and an accompanying app to record data.

According to the researchers, the skin-touching side of the adhesive film collects and analyses the chemical makeup of droplets of sweat. The watch-facing side turns those chemical signals into electrical ones that can be read, processed, and then displayed on the smartwatch.

While the team designed a custom smartwatch and app to work with the system, the research team says the concept could someday be applied to popular models of smartwatches.

Google now shows Covid-19 testing centres in India

Google has partnered with the Indian Council of Medical Research (ICMR) and MyGov to help people find Covid-19 testing centres in India. The tech company has introduced a new testing tab on its Covid-19 related search. This tab provides the location of listed testing centres close to the user. Along with the location, it also shows key



Google has introduced a new testing tab on its Covid-19 related search, which provides the location of listed testing centres (Credit: Google)

information and guidance needed before using their services.

The listing also shows the government-mandated requirements, which include calling the national or state helplines before heading out to get tested, carrying a doctor's prescription (referral required), testing restrictions (tests are limited to certain patients) and information about whether the lab is government or private.

Tapping the 'learn more' link leads to more information from the Ministry of Health and Family Welfare (MoHFW). According to the company, currently, the search has over 700 testing labs listed on Search, Assistant, and Maps spanning more than 300 cities. The new feature is currently available in English, Hindi, Bengali, Telugu, Tamil, Malayalam, Kannada, Marathi, and Gujarati.

Low-cost emergency ventilator to help fight Covid-19 crisis

Fitbit has developed a low-cost ventilator to help save lives during the Covid-19 pandemic. The ventilator, called Fitbit Flow, is a low-cost, high-quality, and easy-to-use automatic resuscitator, developed with the objective of providing ventilation during any emergency.

Fitbit has received a nod from the Food and Drug Administration (FDA) for using the product during the Covid-19 pandemic in situations where conventional ventilators are not available. Inspired by the MIT E-Vent Design Toolbox, the ventilator has been developed based on the inputs from clinicians looking after Covid-19 patients at Oregon Health and Science Hospital and Mass General Brigham Center.

It is easy to use and doesn't require trained professionals to operate. This enables more volunteers to monitor a patient, given that trained staff supply is strained at the moment. The mechanical ventilator is compact and made from laser-cut aluminium. It can sit on a tabletop or a stand with wheels for ease in mobility.

This stand is big enough to fit various sizes of oxygen tanks.

The team at Fitbit is working on improving Fitbit Flow by including a high-resolution add-on display, for viewing the system's status and pressure waveforms.



Fitbit Flow (Credit: Fitbit)